(a) Spaghetti

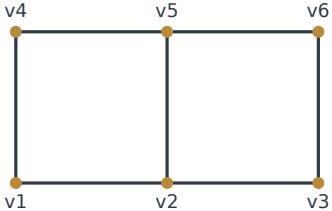
Separate geometry for A and B
2 copies of the shared edge

8 stored segments

Shared edge drawn offset to show both copies; both have x = 1.

(b) Network

Nodes and edges; no faces assigned

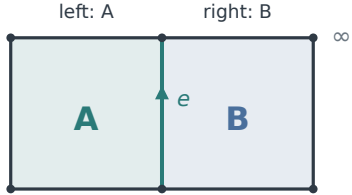


V = 6; E = 7

Nodes record edge connectivity; left and right faces are not stored.

(c) Topology

One edge, two incident faces



V = 6; E = 7; F = 3

F counts A, B and the outer face. Edge e runs from (1, 0) to (1, 1).

Connected planar partition: $V - E + F = 6 - 7 + 3 = 2$